

From samples to Ramachandran contours: Delaunay density, guarded alpha shapes, and a stable density window

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Abstract

A Ramachandran contour turns a cloud of backbone angles into a region that can be read and compared. We describe a triangle-based construction that pursues the same purpose as the published Top8000 kernel-density contours, while retaining a more angular boundary. The method first estimates local density from a periodic Delaunay triangulation and chooses whole triangles to cover the requested fraction of construction observations. A common alpha rule then removes large triangles across sparse gaps, subject to explicit coverage and connectivity safeguards. Finally, a density-window operation removes weak islands and fills eligible shallow holes. Its strength is the first value whose output remains unchanged over a factor of 1.25 at two contour levels and two grid resolutions. The selection uses the triangle field itself; published KDE contours enter only in subsequent assessment. Across six Top8000 classes at the chosen 98% and 99.5% targets, the resulting regions recover the principal reference shapes, with Jaccard scores of 0.712–0.880. The CisPro bridge is removed and fragmentation is greatly reduced, although some islands, loops, and boundary offsets remain different. We explain both the geometric guarantees and the empirical choices, including why local stability is useful without being a proof of the underlying population topology.

1 What should a contour tell us?

The backbone of a protein can be described, residue by residue, by two rotation angles, ϕ and ψ . Plotting these pairs gives a Ramachandran plot. The observations are strongly concentrated: some combinations of angles recur frequently, while large parts of the angular domain are rarely occupied. A useful contour makes this structure visible. It should surround the main concentrations, retain meaningful separate regions, and avoid turning every small fluctuation of sampling into a separate island.

There are two questions hidden in this task. The first is quantitative: *how much of the observed population should the region include?* The second is geometric: *what shape should that region have?* A prescribed percentage answers the first question, but leaves considerable freedom in the second. A thin connection can join two otherwise separate regions while changing very little area. Conversely, a small island may contribute little area but represent a recognizable family of conformations. For this reason, we judge the results by their appearance, area overlap, and numbers of connected pieces and loops, rather than by any one number alone.

Our reference is the six-class Top8000 contour system distributed by the Richardson Laboratory, within the MolProbity validation framework [1, 2]. The classes distinguish glycine (Gly), trans-proline (TransPro), cis-proline (CisPro), residues preceding proline (PrePro), isoleucine and valine

(IleVal), and the remaining General residues. These distinctions matter because their angular distributions differ. In assigning the mutually exclusive classes, Gly and Pro take precedence over PrePro, which takes precedence over IleVal and General [3].

The purpose of the construction here is to obtain a comparable description from local triangulation geometry. It need not produce a smooth curve: straight triangle edges are part of its intended representation. What would make it worthwhile is a readable reconstruction of the main reference regions, a reproducible way to handle sparse connections and small features, and a parameter-selection procedure that can run when no KDE contour is available. Agreement with KDE is evidence about this application, not a definition of the answer on every future dataset.

Figure 1 gives the entire procedure before we develop its parts. Each stage addresses a different difficulty. Local density handles uneven sampling; alpha handles large triangles spanning sparse space; and the final window handles small disconnected islands and shallow holes. None of these steps smooths the density field.

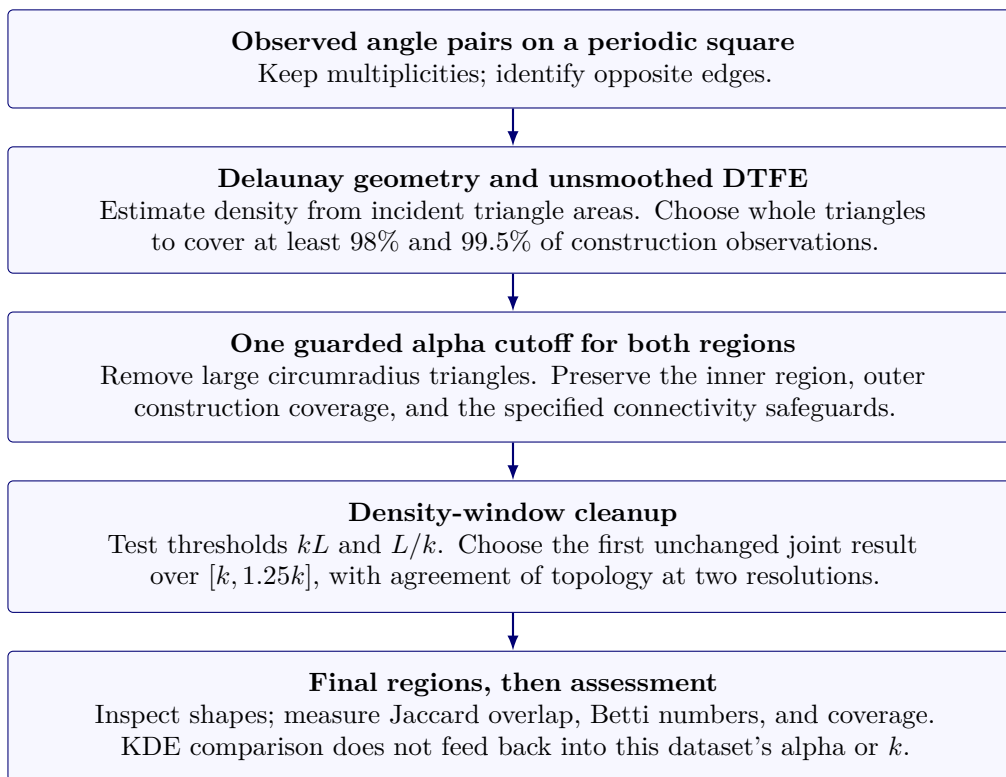


Figure 1: The adopted procedure. The density thresholds L are fixed before alpha and cleanup. The number 1.25 is the required span of stability, not the cleanup ratio k . One rule is used for all six classes, with alpha and k computed from each class's observations.

2 The reference construction and the angular domain

2.1 How KDE turns observations into contours

Kernel density estimation begins by placing a small, smooth mound around each observation and adding the mounds. Where many observations are nearby, their contributions accumulate. The width of a mound determines how much neighboring observations influence one another. An adaptive

KDE varies this width with local sampling: dense regions receive finer detail, while sparse regions are estimated over a wider neighborhood.

The Ramachandran procedure described by Lovell et al. [4] uses two passes with compact radial cosine kernels. A first density estimate supplies the local information used to choose the second-pass widths. The published Top8000 generation recipe specifies periodic evaluation on a 2° grid, with a separate width setting for General and a common setting for the other classes [5]. The final percentage convention ranks density values evaluated at the construction observations. Thus a nominal 98% contour is tied to a fraction of those observations; it is not, merely by its label, a guarantee about previously unseen proteins.

For comparison we use the distributed percentile grids themselves. They store density ranks, rather than probability-density values. We interpolate these scores periodically and bilinearly, and take scores at least 0.02 and 0.005 for our two chosen comparison levels, 98% and 99.5%. The latter is a level selected for this study, not a claim about the standard MolProbity allowed-region cutoff. The interpolation rule and these cutoffs completely specify the reference regions used here. No refitting of the reference is needed to draw the comparisons.

Our method uses construction counts too, but assigns those counts to a different geometric family: unions of triangles. This keeps the purpose of the percentage labels comparable while allowing the density estimator and boundary representation to differ. It also makes an honest distinction possible between achieving a construction target and subsequently measuring agreement with a published region.

2.2 A square whose opposite edges meet

Both angles repeat every 360° . The natural domain is therefore

$$\Omega = (\mathbb{R}/360\mathbb{Z})^2,$$

shown as $[-180^\circ, 180^\circ)^2$. A point near the right edge is close to one near the left edge; a region that crosses an edge has not thereby split into two. The same applies at the top and bottom. Distances use the nearest periodic displacement in each coordinate, followed by the ordinary Euclidean norm, with both coordinates measured in degrees.

The construction identifies equal periodic coordinates and records their positive integer multiplicities w_i . Write x_i for the resulting distinct locations and $N = \sum_i w_i$ for the number of observations. Translated copies of a location may be needed to construct triangles across an edge. They are geometric copies only: their observations are counted once. This convention is used throughout the density calculation, alpha selection, and topology measurement.

3 From Delaunay triangles to a density-based region

3.1 Estimating density from the space around an observation

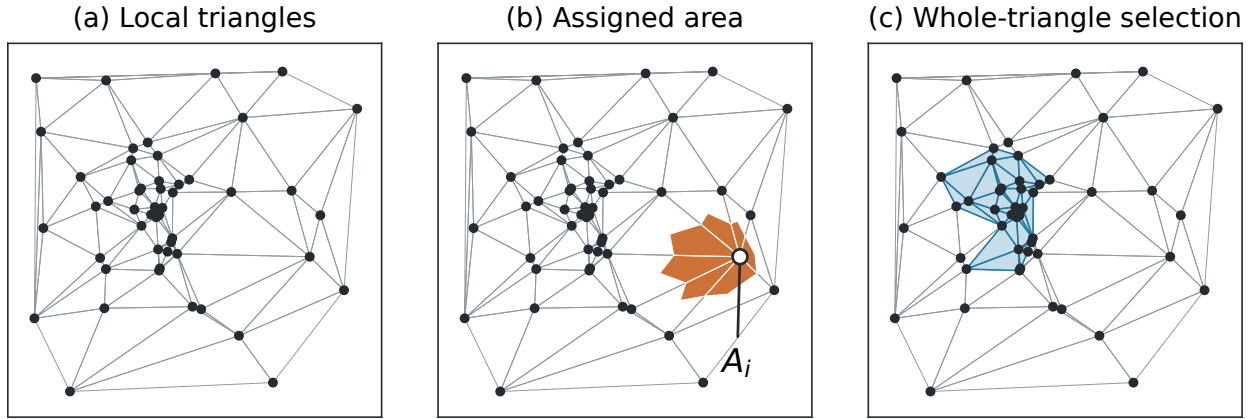
A Delaunay triangulation joins the locations into triangles whose open circumdisks contain no other sample locations. A circumdisk is the disk bounded by the circle through the three corners of a triangle. On our periodic domain, the emptiness condition includes the translated sample locations. The triangulation must cover the angular domain consistently, including its seams. We use a fixed triangulation when a degenerate configuration permits more than one choice.

The triangulation records local sampling scale without first choosing a smoothing width. Nearby observations generally form small triangles; sparse neighborhoods require larger ones. To turn this

geometry into a density, give each corner one third of every incident triangle’s area. The total area assigned to location x_i is

$$A_i = \frac{1}{3} \sum_{\text{triangle corners at } i} \text{area}(T), \quad \rho_i = \frac{w_i}{A_i}. \quad (1)$$

All periodic corner incidences are included. We require $A_i > 0$. In words, the density is the number of observations at a location divided by the area assigned to that location. This is the idea behind the Delaunay Tessellation Field Estimator, or DTFE [6]. Figure 2 illustrates the assignment. The shaded thirds are an accounting device; the algorithm does not need to construct those smaller polygons.



Schematic only: $\rho_i = 1/A_i$; a whole triangle passes when $\min_i \rho_i \geq L$.

Figure 2: A schematic of the density construction. (a) The triangulation adapts to the spacing of the observations. (b) The marked location receives one third of each incident triangle, giving its assigned area A_i . (c) A triangle is retained only when all three corner densities pass the chosen threshold. These illustrative points are not a Top8000 subsample.

There is a useful accounting identity behind this estimate. Interpolate the corner densities affinely inside every triangle, meaning by the unique linear plane taking those three corner values. Call the resulting continuous field $\rho(x)$. The average of an affine function over a triangle is the average of its three corner values, so

$$\int_{\Omega} \rho(x) dx = \sum_T \frac{\text{area}(T)}{3} \sum_{i \in T} \rho_i = \sum_i A_i \rho_i = \sum_i w_i = N. \quad (2)$$

Consequently ρ/N has probability-density units. Dividing all densities and thresholds by N changes no region in our method. No area floor or smoothing correction enters (1).

3.2 Why use whole triangles?

The affine field explains the density estimate and its mass identity, but our contours are built from whole triangles. Give a triangle the score

$$g_T = \min_{i \in T} \rho_i. \quad (3)$$

At a threshold L , retain T precisely when $g_T \geq L$. All its corners then support its inclusion. This is a deliberately conservative way to use the local density: a single high corner does not admit a thin wedge of an otherwise weak triangle. The boundary follows existing edges, which gives it an angular appearance and a clear relationship to the observations.

The minimum-corner score is a convenient selection score, not another normalized density estimator. In particular, integrating this piecewise constant score does not give the mass identity (2). We determine the percentage labels by counting observations in the triangle union.

At this stage there is no alpha restriction. An observation at vertex i belongs to the union if at least one incident triangle is selected. Its inclusion height is therefore

$$h_i(\infty) = \max_{T \ni i} g_T. \quad (4)$$

For $q \in \{0.98, 0.995\}$, sort these heights from largest to smallest and accumulate their integer weights. Choose the highest threshold L_q whose inclusive count reaches $m_q = \lceil qN \rceil$. Include the entire tied group at the threshold; record any resulting excess over m_q . The starting region is

$$S_q(\infty) = \bigcup_{g_T \geq L_q} T. \quad (5)$$

Triangles are closed: their edges and corners belong to the region. An observation incident to several selected triangles is still counted only once, with its multiplicity. This convention also means that two triangles meeting at a single corner are connected.

The 98% region is contained in the 99.5% region because $L_{.98} \geq L_{.995}$. We refer to them as the inner and outer regions. Their percentages are construction targets, achieved before the final cleanup. Equation (5) does not establish coverage on new observations, nor does it assert that the region contains q of the integrated density in (2).

4 Alpha: removing sparse connections without sacrificing the core

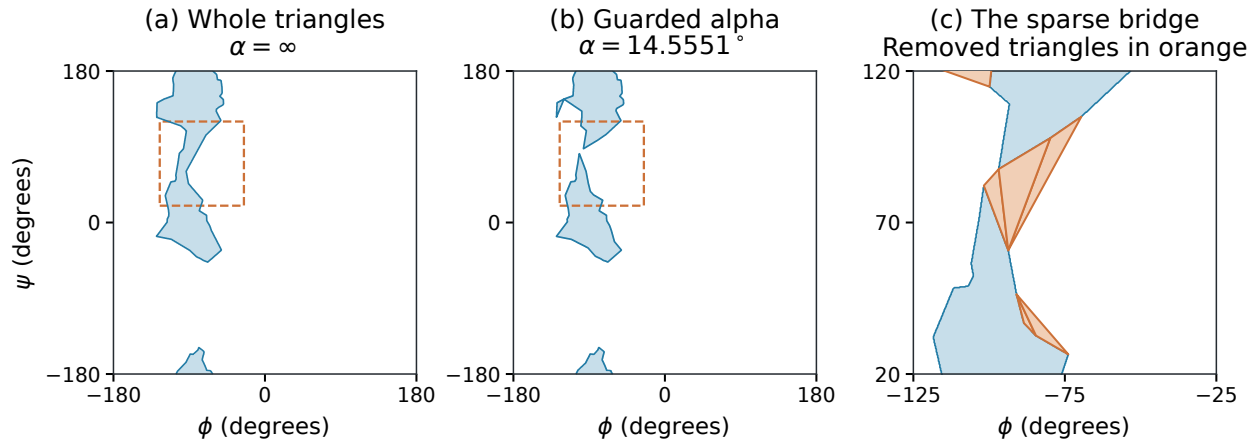
4.1 The geometric problem

Passing a density threshold does not ensure that a triangle provides a plausible connection. Its corners may lie in well-supported neighborhoods while its interior spans a sparsely sampled gap. Whole-triangle selection can then connect concentrations that should remain separate. CisPro gives a concrete example in Figure 3.

Alpha shapes supply a geometric way to limit such triangles [7]. Let R_T be the circumradius of T . For a chosen radius a , we retain only triangles with $R_T \leq a$:

$$S_q(a) = \bigcup_{\substack{g_T \geq L_q \\ R_T \leq a}} T. \quad (6)$$

Here alpha is the radius itself, measured in degrees, rather than its square. We use the filled triangle part of this construction; isolated edges or vertices are not added as separate contour regions. A large circumradius indicates an empty disk at a large angular scale. It can therefore detect both a long span and an unusually thin triangle. The cutoff addresses geometric support, not density averaging.



CisPro, 99.5% construction target. No smoothing; density threshold held fixed.

Figure 3: What alpha changes in CisPro at the 99.5% target. (a) Whole triangles at infinite alpha contain a sparse connection. (b) The common guarded rule selects $\alpha = 14.5551^\circ$ and removes it. (c) The boxed area is enlarged; orange shows the removed triangles in that neighborhood. Eleven triangles are removed in total, reducing the exact outer area by 11.53%. The inner triangles and the required 3,506 outer observations remain. These panels show our geometry alone, without a KDE overlay.

4.2 Choosing one radius by explicit safeguards

A very small radius could destroy useful regions along with an unwanted bridge. We therefore select alpha subject to four safeguards, using the same radius for both levels and holding $L_{.98}$ and $L_{.995}$ fixed. Each safeguard has a particular purpose:

1. **Preserve the complete inner region.** Every triangle in $S_{.98}(\infty)$ must remain. The outer region may be pared back, but not by sacrificing the previously selected inner geometry.
2. **Retain outer construction coverage.** The remaining outer region must still contain at least $\lceil .995N \rceil$ weighted observations. Any removal of observations is limited to the available rounding or tie excess.
3. **Do not create a new isolated excluded region.** Every connected component of the candidate's open complement must contain a triangle from the original outer complement. In ordinary planar language, this prevents excising a new hole from the interior. The condition tracks where excluded space came from, rather than merely comparing total component counts.
4. **Allow only supported splits.** If a connected component of the original outer region splits into two or more surviving pieces, every piece must contain an unchanged inner triangle. Thus a split can separate supported concentrations, but cannot shed new pieces lacking an inner core. An already coreless outer component may remain or shrink; the rule does not require it to disappear.

Connectivity is computed on the periodic triangle geometry. Selected closed triangles connect through shared corners as well as edges. The open complement connects through shared edges of unselected triangles, not by passing through a selected corner. These different conventions follow from the distinction between a closed region and its open complement.

Among candidates passing all four conditions, choose the one with smallest outer area, summing triangle areas before any rasterization. Since the inner region is fixed, this is also the minimizer of the sum of inner and outer areas. Only the active outer-triangle circumradius values and infinity

need be examined: between them the selected triangle sets do not change. All triangles tied at a radius are included. If evaluated candidates give identical triangle sets, prefer the least restrictive of them, with infinity preferred when it gives the same geometry. Infinity always passes, so the method never has to impose a damaging finite cutoff.

This rule does not request two components for CisPro or specify a desired number for any other class. It also does not force different inner components to become different outer components: a supported outer bridge can remain, as in TransPro. Its objective is to remove unnecessary area while respecting the declared structural commitments.

Proposition 1 (Construction thresholds remain calibrated). *If a selected alpha retains the required construction count at L_q , then L_q remains the highest feasible density threshold for that alpha.*

Proof. Every threshold above L_q already covered too few observations at infinite alpha. Imposing a finite alpha only removes triangles, so cannot increase that count. The selected alpha still meets the target at L_q itself. $\square$

Nesting also survives because the same radius is applied to both ordered thresholds. These are properties of the alpha stage. They do not promise that every small island should be retained indefinitely, or that the final cleanup will leave construction coverage unchanged.

5 A density window for islands and holes

5.1 What alpha leaves behind

Alpha can remove a sparse connection, but it need not remove a tiny island made entirely of small triangles. Raw local density estimates can contain many such features, as well as small low-density pockets inside larger regions. Their total area may be modest while their visual and topological effect is substantial. The first column of Figure 4 shows this distinction: General’s broad regions are recognizable, but hundreds of components remain.

Our final step asks whether these features are substantial relative to their own contour’s density threshold. An island should contain a higher density core if it is to survive stronger cleanup. A hole may be filled when it is only a shallow depression, provided filling it does not undo alpha or alter the periodic exterior. This changes entire features while preserving the edge geometry of surviving boundaries; it does not move observations or apply a smoothing kernel.

5.2 Defining the sampled field and its topology

Let D_a be the union of all alpha-eligible triangles, including those below the two contour thresholds. Within this domain, use the score

$$F_a(x) = \max\{g_T : x \in T, R_T \leq a\}, \quad (7)$$

and set it to zero where no eligible triangle contains x . The maximum specifies membership consistently at shared closed edges and corners. Thus $S_q(a) = \{x \in D_a : F_a(x) \geq L_q\}$.

The implemented cleanup uses a square grid. Divide each angular axis into n equal intervals, producing n^2 little squares, which we call pixels or grid squares. Sample (7) and alpha eligibility at each square’s center. A foreground square is one whose center belongs to $S_q(a)$; all other squares form the background. The resulting Boolean array is its mask. We use $n = 768$ and 1536 , with widths 0.46875° and 0.234375° , respectively, and the same origin at -180° .

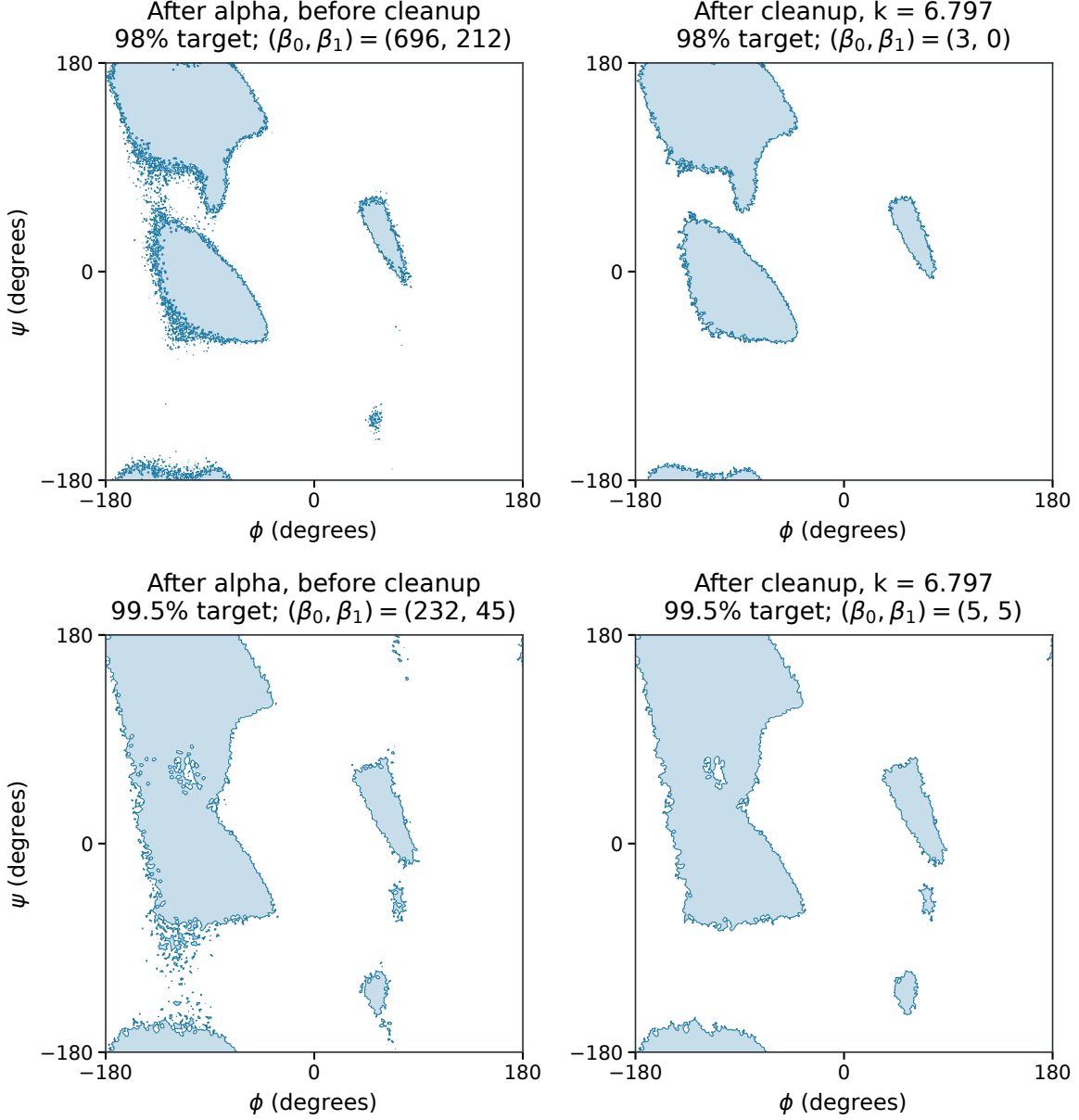


Figure 4: General before and after the selected density-window cleanup. The levels are shown in separate rows to keep their boundaries readable. The 98% component count falls from 696 to 3 and the 99.5% count from 232 to 5 on the 1536×1536 grid. The surviving boundaries remain jagged. Here (β_0, β_1) denotes components and independent loops; the periodic interpretation is explained in the text. No KDE is drawn in this method illustration.

Foreground squares connect along either an edge or a corner (eight neighbors); background squares connect along edges (four neighbors). Opposite grid edges are identified. This gives the topology of a union of closed foreground squares. It approximates the triangle region, but is not a certificate of that region's exact topology.

We record two Betti numbers. The number β_0 counts connected pieces. The number β_1 counts independent loops: closed paths that cannot be contracted within the region. For ordinary bounded patches these often correspond to holes. On a periodic square, a path may also wind around an angular direction, so “loop” and “interior hole” are not synonymous. In particular, opposite-edge fragments must not be counted separately just because the display cuts through them.

5.3 The cleanup operation at a fixed strength

Write L for either L_q and let $k \geq 1$. The two comparison thresholds are

$$L_{\text{high}} = kL, \quad L_{\text{low}} = L/k. \quad (8)$$

They form a symmetric interval on a logarithmic density scale: $[\log L - \log k, \log L + \log k]$. This is the precise meaning of the “epsilon jiggle” here. The level is varied; coordinates are never jittered. Equivalently, $\varepsilon = \log k$ is the half-width in log density. It is not an additive tolerance in degrees.

For each starting foreground component, find the largest sampled score within it, its *peak*. Delete the entire component only if this peak is strictly below kL . Equality retains it. A surviving island therefore contains a core that reaches the higher threshold. This criterion measures relative density support, not island area.

Next recompute the background components *after these deletions*. Removing an island or a weak ring can join background pieces that were previously separate. Using the old background labels could fill the center of a deleted ring as an isolated new island, defeating the operation's purpose. For each recomputed background component, find its smallest score, its *pit*. Fill the component if its pit is at least L/k , except when any of the following applies:

1. it winds around either periodic angular direction;
2. it is tied for largest background area;
3. it contains any alpha-ineligible grid square.

The first two exclusions protect exterior-like background; on a torus there is no unique unbounded exterior to select automatically. All largest-area ties are protected so array traversal cannot decide the answer. The third exclusion protects the *entire* background component containing an alpha-excluded square. In particular, cleanup cannot fill an eligible portion of a gap in a way that restores a connection removed by alpha.

Proposition 2 (A bounded density window). *On the sampled grid, the cleaned region $C_q(k)$ satisfies*

$$D_a \cap \{F_a \geq kL_q\} \subseteq C_q(k) \subseteq D_a \cap \{F_a \geq L_q/k\}. \quad (9)$$

At $k = 1$ the cleanup leaves the starting mask unchanged.

Proof. A square reaching kL_q lies in a starting component whose peak reaches that threshold, so its component is retained. Any retained starting square has score at least $L_q \geq L_q/k$. Any filled square belongs to an alpha-eligible component whose minimum score is at least L_q/k . For $k = 1$, no starting component fails its own level and every unprotected background component contains a square below it, so neither operation changes the mask. $\square$

This gives an explicit spatial and density constraint, not a guarantee that each removed feature was noise. A large but low-peaked island may disappear before a tiny but high-peaked one. Nor can this operation sever a bridge within a retained component: bridge removal belongs to α . Because deletion and filling have different effects, the cleaned regions are not assumed to vary monotonically with k , or to remain nested across the two target levels without checking.

The use of two nearby density levels is related to the approach of Bobrowski et al. [8], who study the topology surviving the inclusion from a higher-level set into a lower-level set. An induced homology map records which components and loops survive that inclusion; its image is their output, rather than a unique cleaned contour. Our whole-component deletion and protected hole filling are a separately specified geometric rule. Raw DTFE, α restriction, this multiplicative window, and the periodic raster do not inherit their kernel-estimation consistency theorem. Equation (9) is a property of our operation; it must not be substituted for that statistical theorem.

A computational precedent for simplifying unsmoothed DTFE topology is DisPerSE [9]. It constructs a discrete Morse complex from Delaunay-based density estimates and cancels persistence pairs below a significance threshold calibrated on Poisson samples. A pair’s density ratio measures its contrast; taking logarithms converts this ratio into an additive difference. Thus DTFE followed by topological simplification, including relative density comparisons, is established prior work. Our operation shares the aim of suppressing weak density features without convolution, but uses the peaks and pits of components at a fixed contour level, with α and periodic-background protections. It neither computes persistence pairs nor cancels critical simplices. In particular, k compares each peak or pit with L ; it is not a persistence-pair ratio. The plateau rule below selects k by output stability, not a Poisson-noise significance calibration, and does not inherit DisPerSE’s significance interpretation.

5.4 Choose the first stable window, then stop

The ratio k controls how aggressively we simplify. Choosing it separately by eye for every class would leave the procedure underspecified. We instead examine the full set of changes over the declared range $1 \leq k \leq 64$. On a finite grid, only finitely many comparisons can change: a peak can cross kL , or a pit can cross L/k . Between actual changes the complete output mask is constant. We call such an interval a *plateau*.

Run this event calculation for both target levels on both grids, and merge the four lists of change points. A joint plateau means that each of the four masks is unchanged throughout that interval. We require complete region identity, not merely unchanged Betti numbers: one island could vanish while another appears, leaving the counts unchanged but the shape different.

The adopted stopping rule selects the first joint plateau with left onset k that satisfies all of the following:

1. it contains the whole closed interval $[k, 1.25k]$;
2. β_0 and β_1 agree between resolutions at each target;
3. both regions are nonempty on both grids;
4. the inner mask is contained in the outer mask on each grid.

Return that k for both target levels of the class. Different classes may return different ratios, but use exactly the same test. If the next actual change is at b , the first condition requires $1.25k < b$, because including b would include a changed region. For a final interval only observed through 64, $1.25k \leq 64$ is sufficient; it makes no claim beyond the observed range. If no interval qualifies, report the selection unresolved. Do not substitute a default strength or silently widen the search.

Strict inequalities require care at event boundaries. For example, a peak equal to kL is retained and disappears just above the limiting ratio. The implementation enumerates the actual floating-point transition onsets, taking the first representable value at which the predicate changes. These saved values, rather than rounded display values, define the returned left endpoint. This operational convention makes the first onset unambiguous even when the corresponding real-number interval would be open on the left.

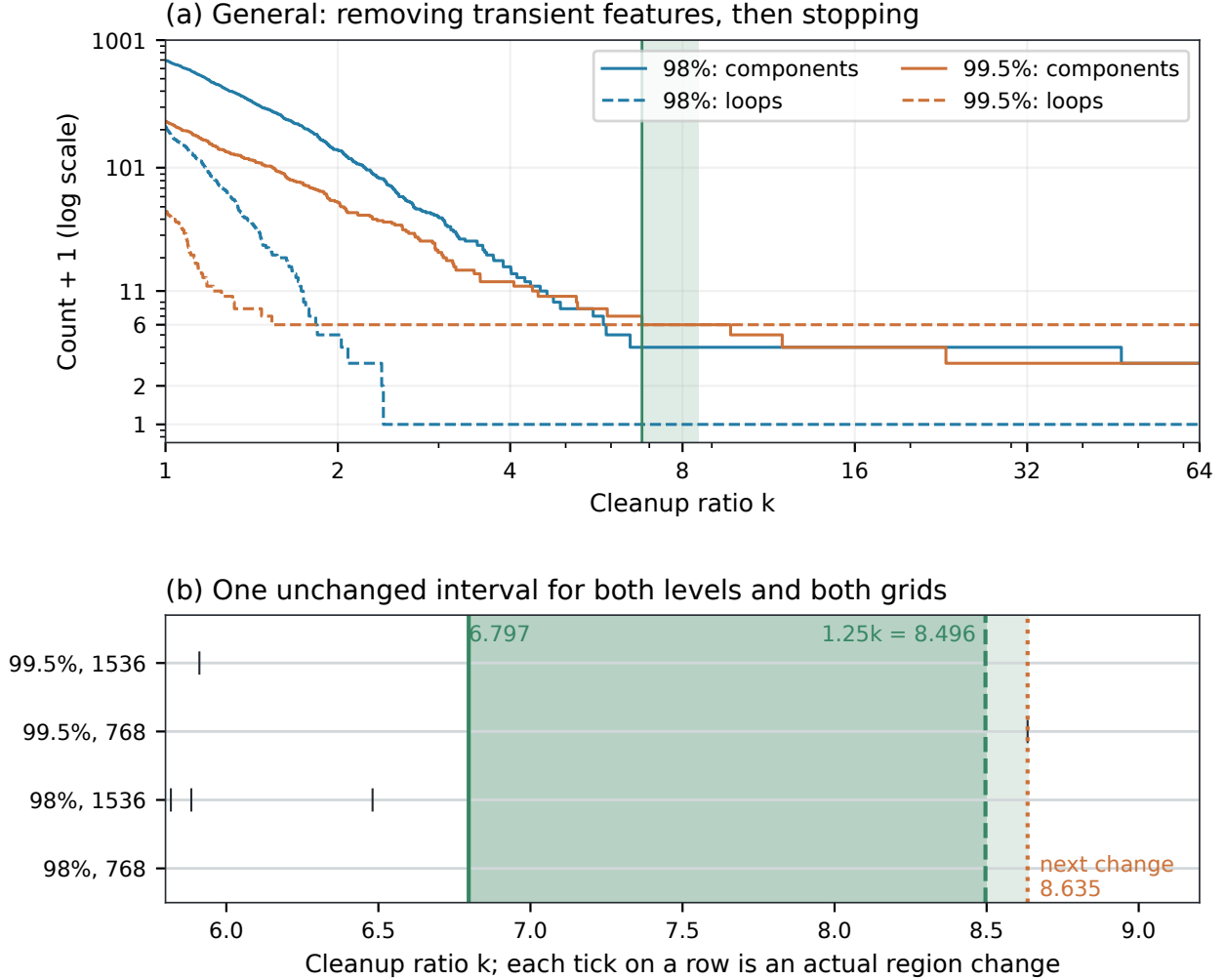


Figure 5: Selecting the General cleanup strength without a KDE target. (a) Components and loops change as k increases; one is added to the counts so zeros can be shown on a logarithmic axis. (b) Each tick records a change of an entire mask. The selected joint plateau begins at $k = 6.796947\dots$ and lasts until $8.634671\dots$. The required interval ends earlier, at $1.25k = 8.496184\dots$, and lies entirely inside it. The displayed Betti curves illustrate simplification; the selector checks mask identity and the additional conditions in Section 5.4.

5.5 Why a factor of 1.25?

The stopping question is local: has a modest further increase in cleanup stopped changing the answer? Requiring a 25% relative increase gives this question a fixed, testable meaning. It is dimensionless, works at $k = 1$ as well as at larger ratios, and, in exact arithmetic, is unchanged by multiplying all density values and thresholds by a common constant. It avoids comparing an

absolute density tolerance across classes with very different populations.

Stopping at the *first* qualifying interval is as important as its width. A stronger cleanup can eventually remove a substantial island and produce another, longer plateau. That does not invalidate the earlier local plateau. If every later simplification were used as a reason to demand more stability, the rule would tend toward increasingly stripped regions rather than a reproducible stopping point.

Figure 6 shows the tradeoff directly. The General outer island near $(73^\circ, -51^\circ)$ survives the selected k , but disappears under stronger cleanup. The small inner island near $(53^\circ, -127^\circ)$ disappears earlier, before the selected interval. The method therefore retains substantial structure without pretending to distinguish every meaningful island from every fluctuation.

The numerical factor 1.25 is an empirical convention for this protocol version, not a uniquely derived constant or a theorem of optimality. It was adopted during Top8000 development with access to the reference comparisons. Its independent justification is the stated local robustness requirement and the preference to stop before further unnecessary feature loss. Once fixed, the per-dataset selector uses only our density field, geometry, and resolution checks. A future application needs no KDE to compute α or k ; a future reference comparison should assess the result without changing this version’s rule. The endpoint 64 and the two grid sizes are likewise declared numerical settings, not universal laws.

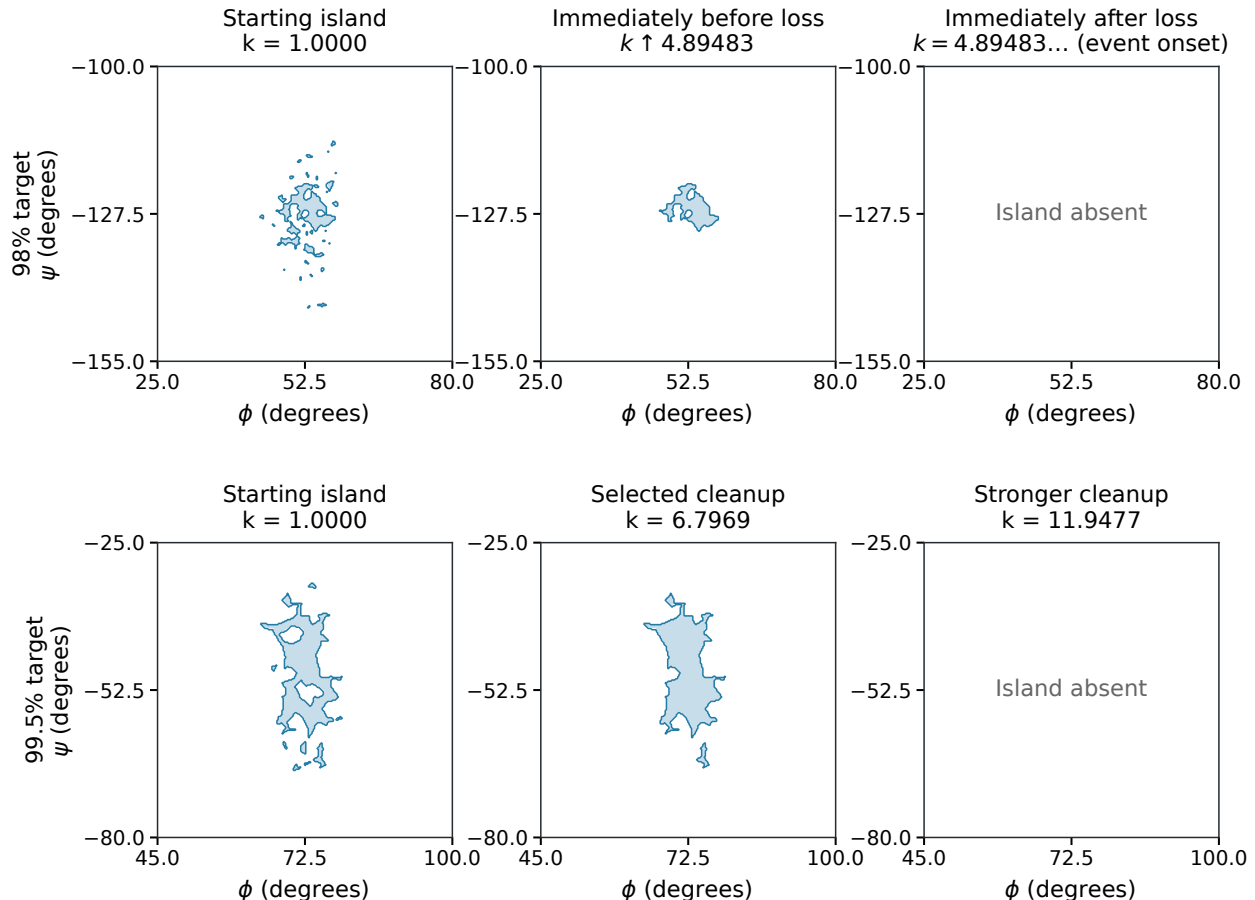


Figure 6: Features have identifiable disappearance thresholds. The top row follows General’s small 98% island through its loss at the event $k = 4.894829 \dots$. The bottom row follows a different, 99.5% island: it remains at the selected $k = 6.796947 \dots$ and disappears at $11.947717 \dots$. Coordinates identify the same neighborhood across each row. These are cropped computed masks, with no KDE input or overlay. Larger k can remove a recognizable island, which explains the decision to stop at the first qualifying plateau.

6 What the six-class results establish

6.1 Parameters produced by the common rule

Table 1 records the full available two-angle construction populations and the computed parameters. No reduced fitting sample is used for these figures. The six numerical α and k values are outputs of the rule, not class constants to copy onto a new dataset. In particular, the larger CisPro radius is not a special instruction for proline.

All six inner triangle regions survive α unchanged. Only CisPro’s outer component splits: its exact triangle topology changes from one component to two, with both pieces retaining inner triangles. Its outer area falls from 9884.44 to 8744.91 square degrees while retaining the required 3,506 observations. For each of the other five classes, less than 0.67% of outer area is removed and outer connectivity is unchanged. These results show why a common guard is preferable to a rule that insists every class should have its outer connections broken.

After α , the first stable choice for CisPro is $k = 1$. Its cleanup is therefore the identity. This is a useful outcome: the same protocol may decide that a class needs no additional simplification. In

Class	Observations N	Alpha radius ($^\circ$)	Cleanup k
General	1,055,696	1.3951	6.7969
Gly	112,542	5.9814	6.0280
TransPro	63,227	2.9527	3.3946
CisPro	3,523	14.5551	1.0000
PrePro	59,498	3.0910	2.8207
IleVal	191,138	2.3693	5.1163

Table 1: Computed outputs for the six Top8000 construction populations. The stability span is 1.25 for every row. Values are rounded for reading; the accompanying numerical record retains full precision for computation.

the denser classes the window removes many small features, while leaving the surviving boundaries at their original angular locations.

6.2 Agreement in area and visible shape

For a final region C and reference region K , the Jaccard score is

$$J(C, K) = \frac{\text{area}(C \cap K)}{\text{area}(C \cup K)}. \quad (10)$$

It equals one for identical regions and zero for disjoint regions of positive area. Our reported values count squares on the common 1536×1536 grid; thus they are raster approximations to area overlap. This is an area measure, not a percentage of observations classified alike.

Figures 7 and 8 show the final regions with the published boundaries. The principal concentrations occupy similar places and have recognizable corresponding outlines. General’s main regions and the broad IleVal and PrePro shapes are recovered; TransPro retains its outer connection; CisPro’s unwanted bridge has been removed. The construction consequently meets the central feasibility goal: Delaunay density and guarded alpha, followed by a reproducible cleanup, can produce readable Ramachandran regions resembling the academic reference without using that reference in per-dataset parameter selection.

The degree of resemblance must still be read quantitatively. Table 2 gives final Jaccard values from 0.712 to 0.880. Cleanup improves overlap for the ten contours it changes and leaves the two CisPro values unchanged. Alpha itself is not an overlap optimizer: it improves CisPro’s outer score from 0.683 to 0.712 while decreasing the other five outer scores by at most 0.0024. Those small costs accompany its explicit geometric objective, rather than a hidden KDE-based choice.

The triangular representation explains much of the finer jaggedness. The raster adds a smaller staircase effect. It does not follow that all disagreement is harmless roughness: some broad boundary offsets and missing islands are real differences between the constructions. Visual comparison is needed precisely to distinguish these cases.

6.3 Topology: substantial simplification, with specific residual differences

The reduction in fragmentation is substantial. For General, the two component counts fall from 696 and 232 to 3 and 5. The selected final Betti pairs agree between the two tested resolutions for every class and level. The inner and outer masks are nested at both resolutions. Seven of the twelve pairs also agree with the reference.

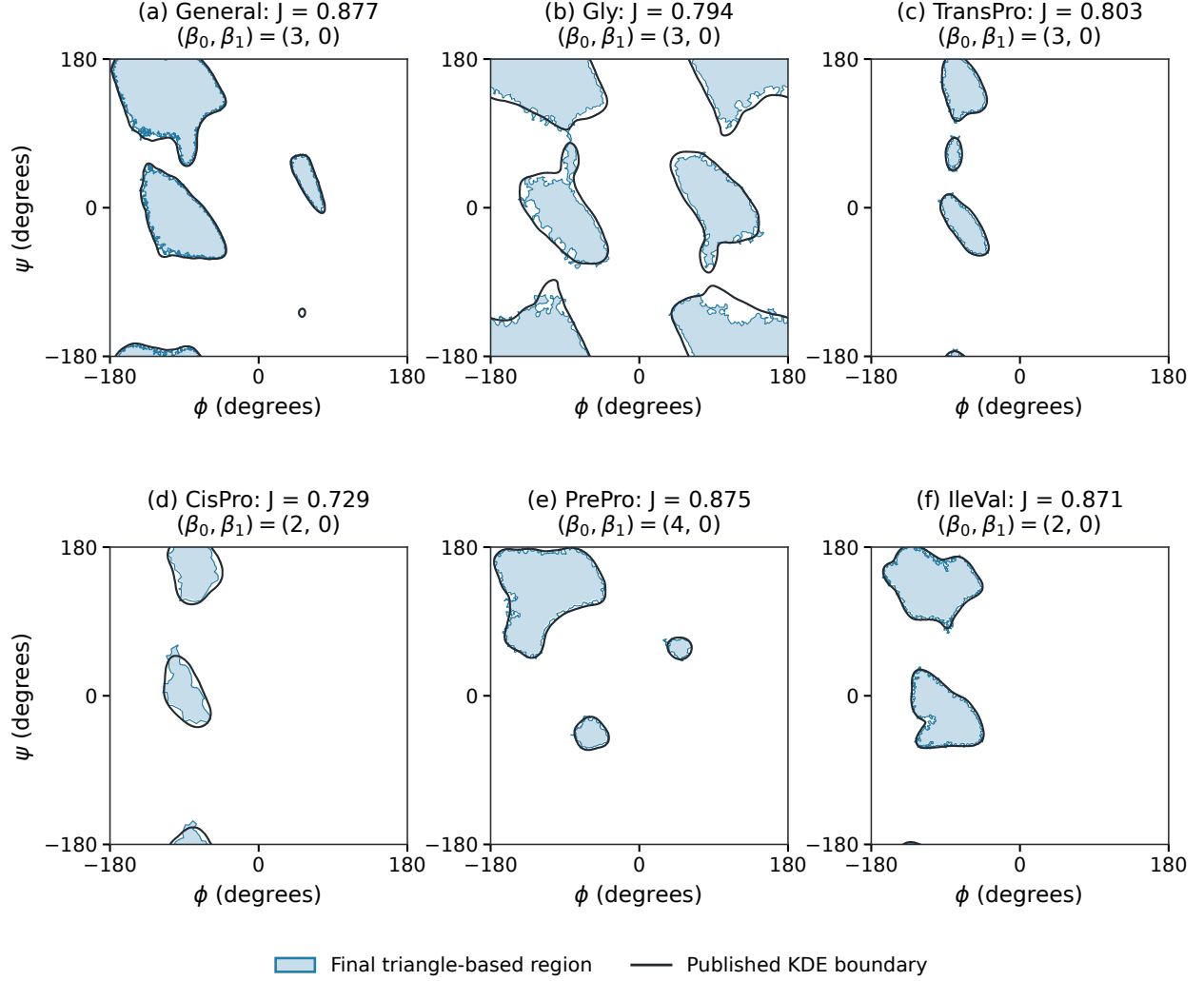


Figure 7: Final regions at the 98% construction target. Blue fill and blue edges show our adopted DTFE-alpha-cleanup result; black lines show the published Top8000 percentile-grid boundary. Each panel reports its Jaccard score and our raster (β_0, β_1) . The same periodic angular domain and display conventions are used in every panel. General’s small lower-right reference island is absent from the final region; PrePro retains an additional component. The label identifies the pre-cleanup construction target, not an exact final raster-coverage statement.

The remaining differences are localized enough to describe. General loses its small inner island, retains five outer components instead of four, and has five outer loops rather than zero. These five loops enclose background connected to alpha-ineligible squares, so the cleanup’s own protection rule deliberately prevents filling them. PrePro retains an extra inner component, and IleVal an extra outer component. CisPro has three outer raster components while the exact selected triangle union has two: a sufficiently narrow contact can disappear when membership is sampled at grid centers. Agreement of the two tested raster counts does not certify the exact contact.

These qualifications matter, but they do not negate the recovery of the principal regions. A Betti number gives the same unit weight to a tiny fragment and a large conformational region; Jaccard can almost ignore a very thin bridge. Their combination with the overlays gives a more useful assessment than optimizing either quantity in isolation. In particular, we do not claim that

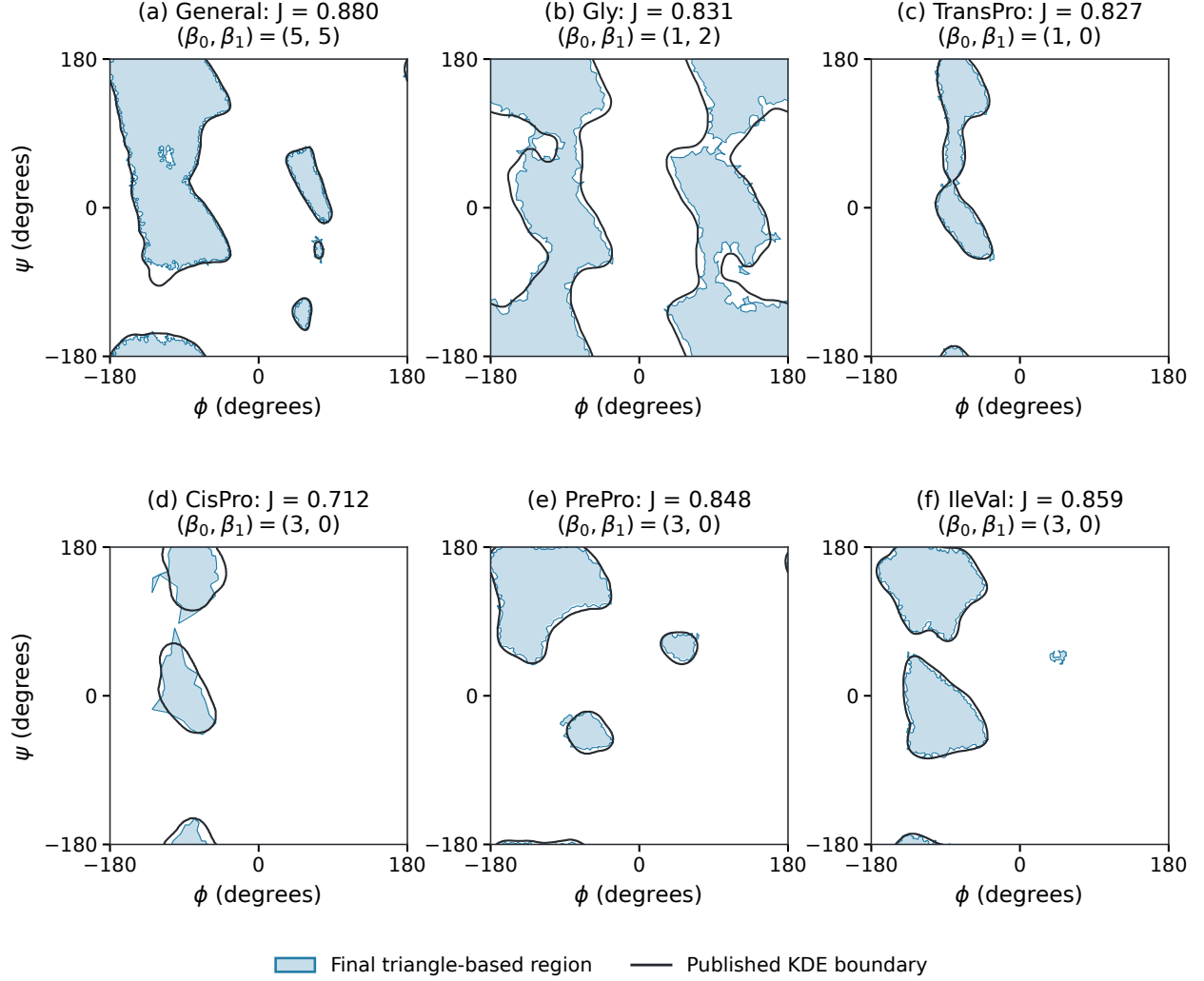


Figure 8: Final regions at the 99.5% construction target, using the same conventions as Figure 7. The dominant shapes remain recognizable, with jagged triangle-derived boundaries. General retains both tracked outer islands, but also retains protected small loops. CisPro has no central bridge; the difference between its exact triangle connectivity and raster connectivity is discussed in the text. IleVal retains a small additional outer component. A reference boundary crossing opposite plot edges remains connected on the angular domain.

matching a Betti pair proves matching topology in location or shape, or that a high overlap score excuses a wrong connection.

6.4 What happens to the percentage labels?

The triangle region after alpha meets its exact construction target. Cleanup then deletes islands and fills holes at the fixed L_q , so it can lower or raise coverage. We measure this change without recalibrating the threshold. The final labels therefore name the construction targets from which the regions were obtained.

There is also a separate numerical effect. The construction counts observations in closed triangles, whereas the cleanup mask classifies an observation by the grid square containing it. A triangle vertex can be inside the exact region while that square’s center is outside. For example, CisPro’s

Class	Target	Jaccard		(β_0, β_1)		
	(%)	Alpha	Final	Alpha	Final	KDE
General	98	0.852	0.877	(696, 212)	(3, 0)	(4, 0)
	99.5	0.871	0.880	(232, 45)	(5, 5)	(4, 0)
Gly	98	0.787	0.794	(71, 9)	(3, 0)	(3, 0)
	99.5	0.823	0.831	(20, 5)	(1, 2)	(1, 2)
TransPro	98	0.792	0.803	(40, 6)	(3, 0)	(3, 0)
	99.5	0.827	0.827	(4, 0)	(1, 0)	(1, 0)
CisPro	98	0.729	0.729	(2, 0)	(2, 0)	(2, 0)
	99.5	0.712	0.712	(3, 0)	(3, 0)	(2, 0)
PrePro	98	0.868	0.875	(25, 6)	(4, 0)	(3, 0)
	99.5	0.847	0.848	(7, 0)	(3, 0)	(3, 0)
IleVal	98	0.845	0.871	(124, 42)	(2, 0)	(2, 0)
	99.5	0.852	0.859	(25, 1)	(3, 0)	(2, 0)

Table 2: Area agreement and topology before and after cleanup, measured on the same 1536×1536 grid. “Alpha” means the guarded alpha result before cleanup. Seven of twelve final Betti pairs match the reference. Counts supplement the overlays: they do not identify the locations, areas, or biological significance of the features.

outer triangle region contains $3506/3523 = 99.517\%$ of its observations, while its raster coverage is 98.325% , even at $k = 1$. This discrepancy is from representation, not an unreported cleanup loss.

Using the same raster convention before and after cleanup isolates the cleanup effect: across the twelve contours the change is between -0.2035 and $+0.0706$ percentage points. For General the two changes are -0.0592 and -0.0421 percentage points. These changes are small in population fraction but do not justify claiming exact final coverage. The numerical record accompanying the paper separates exact starting counts from the two raster counts.

7 A reproducible procedure and its present scope

7.1 What is fixed, and what is learned from a new input?

The protocol fixes the two target fractions, the minimum-corner triangle rule, the four alpha guards, the cleanup operation, the two grids, the search interval, and the 1.25 stability span. Given a new class of angular observations, it computes the Delaunay geometry, the local densities, the two calibrated thresholds, a shared alpha, and a shared cleanup ratio. It can return an unresolved cleanup selection rather than force an answer. This is the sense in which the method stands on its own: the computation does not require a reference contour, a target Betti count, or manually chosen island locations.

Its components have established precedents: DTFE estimates local density from triangulation geometry, alpha shapes constrain empty spatial scales, and density-level comparisons motivate examining the robustness of features. The contribution here is their explicit combination for these angular contour regions, with the interaction between construction coverage, geometric guards, and cleanup spelled out. The application does not require a claim that each ingredient is new.

The implementation records periodic geometry checks, tied observations, calibrated counts, alpha candidates, and complete cleanup change events. For the current data, the periodic mesh exports verify coverage of the angular domain, paired edges, local vertex neighborhoods, and the

required empty-circumdisk conditions. The grid fields use that geometry without smoothing. The selection records are written before the reference assessment. All figures here were regenerated from the adopted saved masks, whose hashes and twelve reported Jaccard values were checked by the accompanying rendering script. The source bundle contains the figure manifest, full-precision numerical tables, and this document’s build recipe.

Computationally, both parameter searches are finite event searches. Two-dimensional periodic triangulations have a number of triangles proportional to their number of vertices; the alpha implementation can skip radii already ruled out by inner preservation or outer coverage. The cleanup can reuse component information and evaluate only actual comparison changes. This is more informative than repeatedly choosing arbitrary trial ratios. It does not by itself establish an end-to-end speed or memory advantage over the academic KDE implementation. The adopted stages have been run on the available six-class inputs; packaging the same stages as a portable new-data interface remains an implementation task, not evidence already supplied by these figures.

7.2 What remains to be established

Top8000 supports feasibility and development of the recipe. It is not an independent validation population after the protocol choices were made. Our supplied construction observations and the distributed reference also need not be identical preparations; notably the published Gly reference is symmetrized, while our fitted observations are not mirrored. Their agreement should be interpreted as agreement of the resulting regional descriptions, rather than exact reconstruction of the reference’s input pipeline.

A prospective Top2018 application should apply this fixed recipe first and compare contours afterward. To interpret coverage on genuinely new observations, overlap or relatedness between the source populations must also be controlled. Within the present data, further numerical work should test boundary convergence and shifted grid origins, rather than treating agreement of two Betti pairs as full convergence. Resampling would address a different question again: whether a feature survives changes to the observations, not merely changes to k .

The intended next coordinate is the side-chain angle χ_1 , with possible later χ_2 and further χ angles. All remain periodic. The DTFE accounting extends naturally: a d -dimensional simplex shares its volume among $d + 1$ corners, yielding the analogue of (2). Delaunay circumradii and nested construction thresholds also have higher-dimensional counterparts. The planar cleanup and its topology safeguards, however, require additional design for three-dimensional tunnels and cavities, and regular grids grow rapidly with dimension. Neither higher-dimensional accuracy nor superior tractability follows from the two-dimensional result. Building and evaluating the (ϕ, ψ, χ_1) extension remains a research objective.

The present conclusion is correspondingly concrete. The three-stage construction produces recognizable, quantitatively similar Top8000 regions with intentionally angular boundaries. It resolves the targeted CisPro connection and most visible fragmentation using the same rules across classes. The remaining islands, protected loops, raster contacts, and coverage qualifications are part of the measured result. They define where the procedure is already useful and where a stronger claim of replacement for the reference still needs evidence.

Data and reproducibility

The reference grids are attributed to the Richardson Laboratory and distributed under CC BY 4.0. We use the pinned Top8000 reference snapshot in the bibliography; the plotted reference boundary is derived by the interpolation and cutoffs in Section 2. Our newly drawn blue regions

use the recorded full six-class construction populations. The numerical procedure is versioned as `dtfe-alpha-epsilon-span125-v1`. The accompanying `README.md` maps each figure and table to its input records and provides commands to rebuild this exposition.

References

- [1] Christopher J. Williams, Jeffrey J. Headd, Nigel W. Moriarty, Michael G. Prisant, Lizbeth L. Videau, Lindsay N. Deis, Vishal Verma, Daniel A. Keedy, Bradley J. Hintze, Vincent B. Chen, Swati Jain, Steven M. Lewis, W. Bryan Arendall, III, Jack Snoeyink, Paul D. Adams, Simon C. Lovell, Jane S. Richardson, and David C. Richardson. MolProbity: More and better reference data for improved all-atom structure validation. *Protein Science*, 27(1):293–315, 2018. doi: 10.1002/pro.3330. URL <https://doi.org/10.1002/pro.3330>.
- [2] Richardson Laboratory. Top8000 Ramachandran percentile contour grids. Versioned data repository snapshot, 2026. URL https://github.com/rlduke/reference_data/tree/cbe300663a2bb55015cf9e9978c720cb12dbf52f/Top8000/Top8000_ramachandran_pct_contour_grids. Repository revision cbe300663a2bb55015cf9e9978c720cb12dbf52f; accessed 22 September 2026. Six rama8000 percentile tables.
- [3] Randy J. Read, Paul D. Adams, W. Bryan Arendall, III, Axel T. Brunger, Paul Emsley, Robbie P. Joosten, Gerard J. Kleywegt, Eugene B. Krissinel, Thomas Lütke, Zbyszek Otwinowski, Anastassis Perrakis, Jane S. Richardson, William H. Sheffler, Janet L. Smith, Ian J. Tickle, Gert Vriend, and Peter H. Zwart. A new generation of crystallographic validation tools for the protein data bank. *Structure*, 19(10):1395–1412, 2011. doi: 10.1016/j.str.2011.08.006. URL <https://pmc.ncbi.nlm.nih.gov/articles/PMC3195755/>.
- [4] Simon C. Lovell, Ian W. Davis, W. Bryan Arendall, III, Paul I. W. de Bakker, J. Michael Word, Michael G. Prisant, Jane S. Richardson, and David C. Richardson. Structure validation by $C\alpha$ geometry: ϕ, ψ and $C\beta$ deviation. *Proteins: Structure, Function, and Genetics*, 50(3):437–450, 2003. doi: 10.1002/prot.10286. URL <https://doi.org/10.1002/prot.10286>.
- [5] Richardson Laboratory. Top8000 filtered-residue generation Makefile. Versioned source repository snapshot, 2026. URL https://github.com/rlduke/reference_data/blob/cbe300663a2bb55015cf9e9978c720cb12dbf52f/Top8000/Top8000_filtered_residues_rotarama/Makefile. Revision cbe300663a2bb55015cf9e9978c720cb12dbf52f; local copy and SHA256 preserved in the reference-provenance directory.
- [6] Willem E. Schaap and Rien van de Weygaert. Continuous fields and discrete samples: Reconstruction through Delaunay tessellations. *Astronomy and Astrophysics*, 363:L29–L32, 2000. URL <https://arxiv.org/abs/astro-ph/0011007>.
- [7] Herbert Edelsbrunner, David G. Kirkpatrick, and Raimund Seidel. On the shape of a set of points in the plane. *IEEE Transactions on Information Theory*, 29(4):551–559, 1983. doi: 10.1109/TIT.1983.1056714. URL <https://doi.org/10.1109/TIT.1983.1056714>.
- [8] Omer Bobrowski, Sayan Mukherjee, and Jonathan E. Taylor. Topological consistency via kernel estimation. *Bernoulli*, 23(1):288–328, 2017. doi: 10.3150/15-BEJ744. URL <https://arxiv.org/abs/1407.5272>.
- [9] Thierry Sousbie. The persistent cosmic web and its filamentary structure—I. theory and implementation. *Monthly Notices of the Royal Astronomical Society*, 414(1):350–383, 2011. doi: 10.1111/j.1365-2966.2011.18394.x. URL <https://doi.org/10.1111/j.1365-2966.2011.18394.x>.